JOHANSSON ENGINEERING

AFS 2017:3 — Consolidated Supervision Requirements

Document No.	JDS-RPT-MEC-003
Revision	B
Date	2026-04-10
Status	CURRENT
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Regulation	AFS 2017:3 consolidated with AFS 2019:1, AFS 2020:10, AFS 2022:2

Disclaimer

This document is a **practical English-language summary** of the Swedish regulations governing ongoing supervision of pressurised equipment. It has been verified against the official PDF texts published by Arbetsmiljöverket. It does not replace the official Swedish texts.

1. Regulation Overview

1.1 Base Regulation

AFS 2017:3 — "Use and Inspection of Pressurised Equipment" — entered into force 1 December 2017, replacing AFS 2005:3 (inspection), AFS 2002:1 (use), AFS 2005:2 (manufacture), and AFS 2001:4 (gas cylinders).

1.2 Amendments

AMENDMENT	IN FORCE	WHAT CHANGED
AFS 2019:1	2 Sept 2019	Textual: 1 Kap. §2 (scope clarified for vehicles), §6 (added "enkla tryckkärl" definition), 3 Kap. title/§1/§5/§7 (CO2 fire extinguishers specified), 5 Kap. §3/§4 (wording), 6 Kap. §11 (inställelsetid: "1 or 2" replaces "1 and 2")
AFS 2020:10	1 Feb 2021	Transitional: punkt 4 changed — boiler operator certification (all categories 1-4) extended to 5 years after entry into force
AFS 2022:2	30 May 2022	Transitional: punkt 4 changed — boiler operator certification deadline set to 1 December 2025

Note: AFS 2020:10 and AFS 2022:2 only changed transitional provisions (övergångsbestämmelser). No substantive requirements in the main regulation were altered by these amendments.

2. Scope of Application (1 Kap.)

2.1 What Is Covered (§2)

The regulation applies to **use and inspection** of pressurised equipment and safety equipment. This includes: - **Pressure vessels** (tryckkärl) — closed vessels for fluids under pressure - **Cisterns** — vessels for liquids where gas pressure does not exceed 0.5 bar above atmospheric - **Vacuum vessels** (vakuumkärl) — vessels below atmospheric by more than 0.0065 bar - **Piping** (rörledning) — pipe systems including hoses, fittings, and armature - **Safety equipment** (säkerhetsutrustning) — devices preventing pressure/temperature from exceeding safe limits

2.2 What Is NOT Covered (§2)

Nuclear installations, aircraft/rockets, military equipment, inflatable toys/boats, blast furnaces, radiators, ventilation ducts, water supply systems, beverage distribution (≤7 bar and ≤500 bar·L), vehicle equipment, electrical generation/distribution equipment.

2.3 Key Definitions (§6)

SWEDISH	ENGLISH	MEANING
Användning	Use	Includes operation, monitoring, ongoing supervision, maintenance, cleaning, storage
Fortlöpande tillsyn	Ongoing supervision	Regular examination to assess whether equipment has been damaged or deteriorated
Övervakning	Monitoring	Task to ensure operation is correct and safe
Säkerhetsutrustning	Safety equipment	Devices preventing pressure/temperature exceeding safe limits
Fluid	Fluid	Gas, liquid, vapour (including suspensions)
Panna	Boiler	Pressure vessel transferring heat from fuel/energy to fluid for external use

3. General Requirements for Use (2 Kap.)

3.1 Risk Assessment (§1)

The employer must assess risks considering: 1. Experience from use of the equipment 2. Information about remaining lifetime 3. Repairs and modifications performed 4. Accidents and incidents that have occurred 5. Deviation reports and inspection results

3.2 Placement (§2-3)

Equipment must be placed so that: - Maintenance, supervision, and inspection can be performed safely - It is protected from damage by nearby activities (e.g., vehicle collision) - Consequences of an accident are minimised (blast zone, access, evacuation) - It is not exposed to temperatures below its design minimum

3.3 Ongoing Supervision — Mandatory Minimum Checks (§6)

This is the core requirement for the supervision program. To remain pressurised, equipment must be regularly examined through ongoing supervision. The supervision shall **at minimum** cover:

1. That the equipment **functions satisfactorily**
2. That **no leaks** have occurred
3. That the equipment has not been subjected to **harmful external or internal impact**
4. That **no other faults or deviations** have occurred
5. That pressurised equipment, valves, and emergency stops are **correctly marked**
6. That any **prescribed inspections** (per 3 Kap., 5 Kap., or 1 Kap. §4) have been carried out

> These six points (§6 punkt 1-6) are the **legal minimum** that every supervision program must address. The program generator uses these as the baseline check requirements.

3.4 Disassembly/Assembly (§7)

Equipment must be depressurised and drained before disassembly. Piping may be connected to pressurised equipment only with risk-assessed procedures describing method, authorised persons, inspection plan, and exclusion zone.

4. Class A and B Equipment (4 Kap.)

4.1 Classification — Pressure Vessels (§10)

Pressure vessels not covered by §7-9 (boilers, water heaters, directly heated vessels) are classified by the table in §10 based on PS × V and fluid group:

Group 1a (dangerous fluids) — gases:

VOLUME	PRESSURE	CLASS
V > 1 L	p > 0.5 bar	**B** if PS×V ≤ 1,000; **A** if PS×V > 1,000
0.1 < V ≤ 1 L	p > 200 bar	**A**

Group 2a (non-dangerous fluids) — gases:

VOLUME	PRESSURE	CLASS
V > 1 L	p > 0.5 bar	**B** if PS×V ≤ 10,000; **A** if PS×V > 10,000
0.1 < V ≤ 1 L	p > 1,000 bar	**A**

Critical exception (§10, paragraphs after table):

> Pressure vessels for **air and nitrogen gas** that would be Class B shall **not belong to any class**. This means air receivers and nitrogen vessels below the Class A threshold are **exempt from periodic inspection** requirements.
>
> Pressure vessels in **refrigeration/heat pump systems** containing Group 2a gas that would be Class B shall **not belong to any class**.
>
> Pressure vessels containing Group 2a **liquid at ≤65°C** do not belong to any class.

4.2 Classification — Boilers (§7)

TEMPERATURE	CLASS
t > 110°C	**A** (if rated power > 5 kW)
65°C < t ≤ 110°C	**B** (if rated power > 100 kW)

Flow-through boilers with coil volume ≤25 L never belong to Class A or B.

4.3 Classification — Piping (§11)

Based on DN × p (nominal diameter × pressure) and fluid group. More restrictive for Group 1a.

4.4 Coordination Person (§14)

The employer must assign a **physical person** to plan and coordinate work when Class A/B equipment is assembled, installed, disassembled, cleaned, serviced, maintained, or inspected. This person must ensure written work instructions exist.

4.5 Equipment Register (§15)

The employer must maintain a **register of all Class A and B equipment** at each workplace.

4.6 Monitoring of Class A and B Vessels (§16)

Class A and B vessels must be **continuously monitored** (operatören omedelbart kan nå behållaren — the operator can immediately reach the vessel). Exception: if the employer can demonstrate through documented risk assessment that periodic monitoring with defined intervals is safe, including provisions for safety-related alarms and response times.

4.7 Documented Supervision Routines (§17) — MANDATORY

The employer who allows Class A or B equipment to remain pressurised **must** have **documented routines for ongoing supervision**. The routines must show how the employer ensures compliance with 2 Kap. §6 points 1-6.

The employer must: - **Evaluate and revise** the routines **at least once per year** - Assign a **physical person** responsible for ensuring supervision is carried out and documented

Sanction fee for non-compliance: 10,000 — 100,000 SEK (calculated per number of employees).

4.8 Lifetime Journal (§18)

The employer must maintain a **journal showing remaining lifetime** for Class A/B equipment with limited lifetime. If different parts have different lifetimes, the journal must describe each part separately.

Equipment that has reached its documented lifetime may only remain pressurised if the employer has conducted and documented an analysis showing extended lifetime.

4.9 Deviation Reports (§19)

Actions taken because Class A/B equipment has been damaged or deteriorated must be documented in a **deviation report** containing:

1. Description of the damage or deterioration
2. How it was discovered
3. Date of discovery
4. What action is needed
5. Cause (if not obvious)
6. Date when action was taken
7. Who made the report

5. Inspection (5 Kap.)

5.1 Condition for Pressurisation (§1)

Class A or B equipment may only be pressurised during a period for which an inspection body has assessed that it meets requirements. It may not be used at pressures or temperatures deviating from the inspection body's assessment.

Sanction fee for pressurising without valid inspection: 15,000 — 300,000 SEK (max 600,000 SEK total).

5.2 First Inspection (§3)

Class A/B equipment must undergo first inspection before initial pressurisation or after relocation (if stationary).

5.3 Recurring Inspection — Overview (§4)

Class A and B equipment must undergo recurring inspection per **Bilaga 1**.

- **Class B:** Recurring inspection = driftprov (operational test) only
- **Class A:** Recurring inspection = driftprov, or driftprov + in- och utvändig undersökning (internal/external examination)

5.4 After Inspection (§18-21)

The inspection body assesses: - Whether the equipment meets requirements - Maximum/minimum pressure and temperature - **Expiry month** for next recurring inspection

A **control plate** (skylt) must be attached to vessels showing identity, expiry month, inspection body mark, and pressure/temperature limits.

An **inspection certificate** (intyg) must be issued after every inspection.

5.5 Timing Window (§22)

Recurring inspection may be performed **6 months before to 5 months after** the expiry month. If the interval is 6 months, inspection must be done before expiry.

6. Recurring Inspection Intervals (Bilaga 1)

6.1 Driftprov (Operational Test) — Base Intervals (§1.4.1)

EQUIPMENT TYPE	INTERVAL
Pressure vessels in refrigeration/heat pump systems	4 years
Vessels filled with air, nitrogen, or noble gas (Group 2a)	4 years
Vacuum-insulated vessels with condensed O2, N2, Ar, CO2	4 years
LPG storage vessels	4 years
Vessels with liquid-phase fluids (thermal expansion only)	4 years
All other pressure vessels and vacuum vessels	**2 years**
Cisterns	3 years
Piping connected to cisterns	3 years
Other piping	4 years

Extended interval (§1.4.2): If safety equipment functioned without action at the two previous operational tests, and the system check could be replaced by reviewing the employer's supervision documentation, the interval may be extended. **Maximum: 4 years.**

Shortened interval (§1.4.3): If safety equipment did NOT function without action, the next functional test interval is **halved**.

6.2 Internal/External Examination — Condition-Based Intervals (§2.2)

This is **NOT a fixed interval**. The inspection body determines the interval based on equipment condition. The system is progressive — longer intervals require increasingly clean inspection history:

INTERVAL	CONDITIONS REQUIRED
4 years	Not fire-affected; remaining lifetime > 5 years; low crack risk material; non-harsh environment
2 years	4-year conditions not met, but: ≥2 prior 1-year inspections or ≥1 prior >2-year inspection; no repairs needed; >3 years before corrosion requires action; remaining life > 3 years
1 year	2-year conditions not met, but safe for 1 year per inspection body and risk assessment
6 months	1-year conditions not met, but safe for 6 months
6 years	Meets 4-year conditions PLUS: ≥1 clean 4-year inspection; no fatigue/creep; safe for 8 years
8 years	Meets 6-year conditions PLUS: ≥2 clean 4/6-year inspections; no corrosion/erosion; safe for 10 years
10 years	Meets 8-year conditions PLUS: ≥2 progressively longer inspections; no degradation; safe for 12 years
12 years	Cisterns only: meets 6-year conditions; ≥1 clean 6-year inspection; safe for 14 years

> The external and internal examination may be split so that external is done more frequently (§2.2.1).

6.3 Exemptions from Internal/External Examination (5 Kap. §6)

The following Class A equipment is **exempt** from internal/external examination:

1. Heat exchangers in district heating substations (≤120°C)
2. Vacuum-insulated vessels with condensed CO2
3. Deep-cooled condensed gas vessels (boiling point < -153°C)
4. Hydraulic oil system vessels (if material has low crack risk)
5. Refrigeration/heat pump vessels (if medium not corrosive to material)
6. Water piping ≤120°C laid in a protected manner

7. Boiler Monitoring (6 Kap.)

7.1 Monitoring Types

- **Continuous monitoring** (ständig övervakning): operator immediately at the vessel (§16, 6 Kap. §6)
- **Periodic monitoring** (periodisk övervakning): allowed only if equipment prevents exceeding predetermined limits (6 Kap. §7)

7.2 Boiler Operator Certification

Boiler operators must be certified by an accredited certification body (6 Kap. §4). Four categories per Bilaga 2. **Certification deadline: 1 December 2025** (per AFS 2022:2).

8. Safety Equipment

Safety equipment must be: - Present and functioning (checked at every driftprov per Bilaga 1 §1.2) - Not defeated or rendered inoperative - Set to correct activation values

If safety equipment did not function at the previous driftprov, the next test interval is **halved** (Bilaga 1 §1.4.3).

9. Sanctions

VIOLATION	FINE RANGE
No documented supervision routines for Class A/B (4 Kap. §17)	10,000 — 100,000 SEK
Pressurising without valid inspection (5 Kap. §1)	15,000 — 300,000 SEK (max 600,000 total)
Operating boiler without valid monitoring assessment (6 Kap. §1-2)	15,000 — 300,000 SEK (max 600,000 total)

10. Accredited Inspection Bodies (7 Kap.)

Inspections must be performed by bodies accredited per EU Regulation 765/2008. In Sweden, accreditation is by SWEDAC.

ORGANISATION	ACCREDITATION
DEKRA Industrial AB	SWEDAC accredited
Kiwa Sweden AB	SWEDAC accredited
RISE Research Institutes of Sweden	SWEDAC accredited
Bureau Veritas Sweden	SWEDAC accredited
TUV Nord Sweden	SWEDAC accredited

11. Swedish Terminology Reference

SWEDISH TERM	ENGLISH TERM (JDS)
Fortlöpande tillsyn	Ongoing supervision
Tillsynsprogram / rutiner	Supervision program / routines
Driftprov	Operational test (recurring inspection)
In- och utvändig undersökning	Internal and external examination
Återkommande kontroll	Recurring inspection
Revisionskontroll	Revision inspection (after repair/change)
Första kontroll	First inspection (before initial pressurisation)
Tryckkärl	Pressure vessel
Trycksatt anordning	Pressurised equipment
Brukare / Arbetsgivare	Operator / Employer
Ackrediterat kontrollorgan	Accredited inspection body
Kontrollintyg	Inspection certificate
Säkerhetsutrustning	Safety equipment
Avvikelserapport	Deviation report
Förfallomånad	Expiry month (for next inspection)
Sanktionsavgift	Sanction fee

Revision History

REV	DATE	AUTHOR	DESCRIPTION
A	2026-04-10	N. Johansson	Initial release — summary based on general knowledge
B	2026-04-10	N. Johansson	Complete rewrite — verified against official PDF texts. Corrected inspection intervals (condition-based per Bilaga 1, not fixed 24/72/144), added air/nitrogen Class B exemption (§10), added deviation reports (§19), lifetime journal (§18), coordination person (§14), monitoring requirements (§16), sanction fees, full Bilaga 1 interval tables